

System Design of Competitive Programming Contest Platform

1. Questions

1.1. Type

1.1.1. MCQ

1.1.2. Coding questions

1.2. Level

1.2.1. Beginner

1.2.2. Intermediate

1.2.3. Hard

1.3. Topic

1.3.1. Introduction

1.3.2. Variables & Data Types

1.3.3. Loops & Conditionals

1.3.4. Arrays & Strings

1.3.5. Functions & Recursion

1.3.6. Sorting & Searching

1.4. Language

1.4.1. C

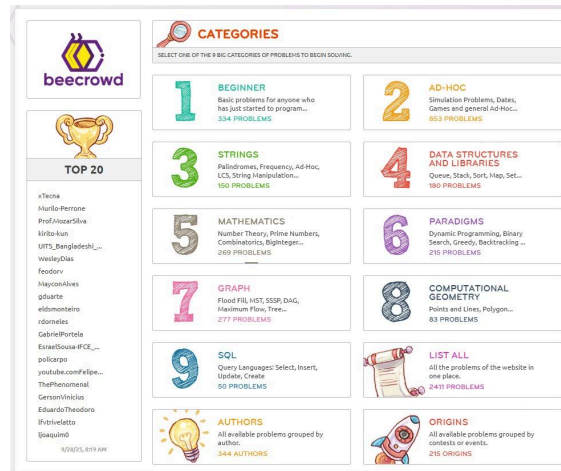
1.4.2. C++

1.4.3. Java

1.4.4. Python

1.4.5. PHP

1.4.6. JavaScript



2. Papers for Practice

2.1. Topic wise

2.2. Level wise

2.3. Language wise

3. Solution Sets

3.1. Question Explanation

3.1.1. Tutorial

3.1.2. Hints

3.1.3. Examples

3.2. Solution Explanation

3.1.1. Tutorial

3.1.2. Code Sample

3.3. Common Mistakes

4. Roadmap

4.1 Beginner Roadmap

4.1.1 Fundamentals

4.1.1.1. Programming basics

- syntax

- variables
- data types
- operators
- conditionals
- loops
- functions.

4.1.1.2.Tools & environment setup

- Install a code editor
- command-line basics
- set up Git/GitHub
- install chosen language runtime

4.1.1.3.First small projects

- Calculator App
- To-Do List

4.1.2 Core Concepts

4.1.2.1. Data structures

- Arrays/lists
- maps

4.1.2.2. Input/output handling

4.1.3 Practice Projects

4.1.3.1. Mini games

- Tic-tac-toe
- number guessing
- rock-paper-scissors

4.1.3.2. Using public APIs

- weather data
- currency exchange rates
- Random quotes generator

4.1.4 Recommended Videos/Documentations

4.1.4.1. Official docs

4.1.4.2. Beginner crash courses

- YouTube

4.1.4.3. Interactive tutorials

- W3Schools
- Codecademy

4.2 Intermediate Roadmap

4.2.1 Skill Expansion

- OOP
- File handling

4.2.2 Applied Knowledge

- Databases
- API (RestAPI)
- Frameworks introduction

4.2.3 Projects

- CRUD app
- Blog/notes app

4.2.4. Recommended Videos/Documentations

- Framework official guides (Laravel, React)
- Intermediate tutorials

4.3 Advanced Roadmap

4.3.1 Advanced Concepts

- Design patterns
- Advanced algorithms

4.3.2 System-Level Skills

- Architecture (MVC)

4.3.3 Advanced Projects

- Full-stack application

4.3.4. Recommended Videos/Documentations

- Advanced system design lectures

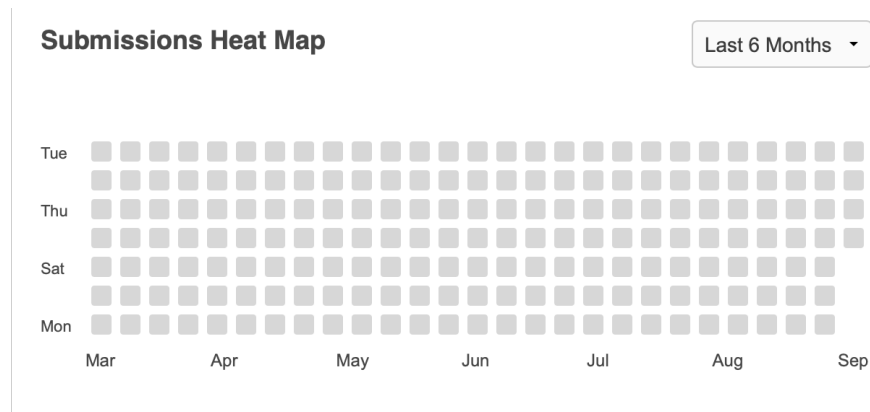
5. Profile

5.1. Personal Info

- 5.1.1. Profile Picture
- 5.1.2. Username
- 5.1.3. Country
- 5.1.4. Student/Professional
- 5.1.5. Gender
- 5.1.6. Email
- 5.1.7. Institution Name
- 5.1.8. Programming Language Preference

5.2. Progress & Statistics

5.2.1. Submissions Heat Map



5.2.2. Learning Paths



5.2.3. Practice Paths



5.2.4. Contests

- List of contests

5.2.5. Total Problems Solved

5.3. Achievements / Badges

5.3.1. Code Enthusiast - one Badge (min 10 problems need to solve)

5.3.2. Code Enthusiast - one Badge (need to attend 5 contests)

5.3.3. Problem Solver - Bronze Badge (Solve 200 Problems)

5.4. Recent Activities

Recent Activity				
Time	Problem	Result	Lang	Solution
🕒	KITCHEN...	✓ (100)	C++	Explain
🕒	RATINGIN...	✓ (100)	C++	Explain
🕒	SEVENRI...	✓ (100)	C++	Explain

Reference : <https://www.codechef.com/>

6. Study Plan

6.1. Daily Plan

- 6.1.1 Warm-up (easy problems or quizzes)
- 6.1.2 Practice (1–3 problems on a topic)
- 6.1.3 Timed drill (solve 1–2 problems like in a contest)
- 6.1.4 Review (check mistakes and read solutions)

6.2. Weekly Plan

- 6.2.1. Focus on 1–2 topics per week
- 6.2.2. Solve mixed problems from different topics
- 6.2.3. Take 1 full mock contest
- 6.2.4. Review unsolved problems

6.3. Monthly Plan

- 6.3.1. Practice key topics and algorithms
- 6.3.2. Do 2–4 mock contests
- 6.3.3. Check strengths and weaknesses
- 6.3.4. Set next month's goals
- 6.3.5. team practice or group projects

6.4. Recommended Tasks

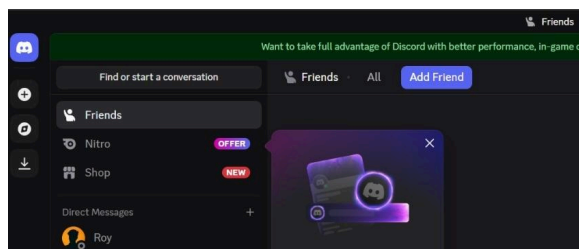
- 6.4.1 Problem Sets
- 6.4.2 Learning Tasks
- 6.4.3 Project Task

6.5. Progress Tracking

7. Community

7.1. Announcement

7.2. Discussion



Reference : <https://discord.com/>

8. Notification

8.1. Reminders

8.1.1. For upcoming contest

8.1.2. Daily Practice

8.1.3. Streak

8.1.4. Study Plan

8.2. Event Notifications

8.2.1. Upcoming Events

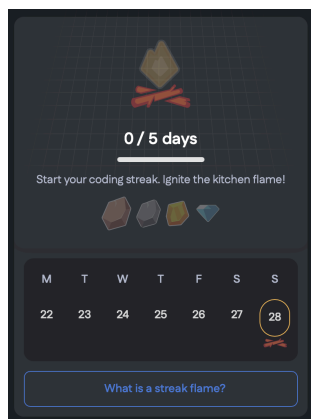
8.2.2. Ongoing Events

8.2.3. Announcements

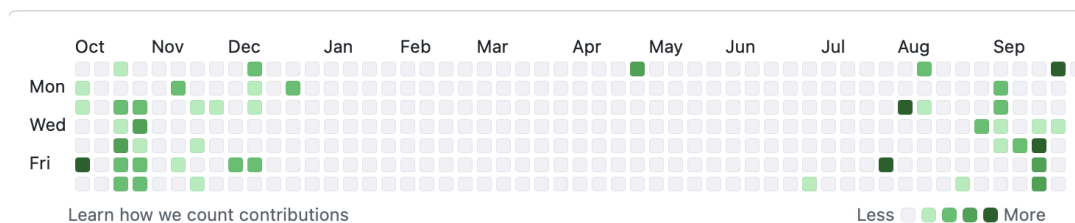
8.3. Achievement / Badge Notifications

9. Streak

9.1. Daily Activity Tracking



9.2 Streak Graph



Reference : [DuoLingo](#) , [GitHub](#) , [Snapchat](#) , [CodeChef](#)

10. Event

10.1. Contest

10.1.1 Coding Contest

10.1.2 Bug-Fixing Contest

10.1.3 Hackathon

10.1.5 Quiz

10.2. Workshop

10.2.1 Beginner Workshops

- Intro to programming languages
- Git & GitHub basics
- Building first projects

10.2.2 Intermediate Workshops

- Web development bootcamps
- API building & integration

10.2.3 Advanced Workshops

- System design & architecture

10.3. Team Challenges